

Electronic Dice Roller

Prangan Barua(23201370)
Dept. of CSE
Brac University
Dhaka, Bangladesh
prangan.barua@g.bracu.ac.bd

Ahmed Rakin(23301039)
Dept. of CSE
Brac University
Dhaka, Bangladesh
ahmed.rakin@g.bracu.ac.bd

Md Moniruzzaman(24101439)
Dept. of CSE
Brac University
Dhaka, Bangladesh
md.moniruzzaman5@g.bracu.ac.bd

Abstract—This project presents the design and implementation of an Electric Dice Roller. The main objective is to replace a traditional physical dice with a digital circuit. When a push button is pressed, the circuit generates a random number between 1 and 6, and displays the result using LEDs. This project demonstrates the practical application of digital electronics and hardware components in an interactive way.

Keywords —Electronic Dice, 555 Timer, Astable Multivibrator, 74161 Counter, NAND Decoding Logic, CD4511 BCD Decoder, Seven-Segment Display, Random Number Generation.

I. Implementation

A. Theory

This project is a design of a Electronic Dice Roller using a combination of combinational and sequential logic ICs without any microcontroller. It includes a free running pulse generator, a synchronous counter whose output is limited to six states by decoding logic, and a BCD to 7-segment driver to signal the output on a display. After pressing the push-button, the circuit will keep the count going, and releasing it will cause it to stop on the value that it reaches when it is released, thus showing the final dice roll.

1) Pulse Generation (NE555 Astable Multivibrator): The NE555 timer is made to free run or produce a continuous square-wave clock at its OUTPUT pin by simply placing a timing resistor and a 0.1 μF capacitor into astable mode. This clock is used for the clock input of the counter stage.

2) Counting (74161 4-bit Synchronous Counter): The 74161 is a synchronous 4-bit binary counter. When the clock goes high, the outputs QA–QD of the 555 timer increment by one – using a 4-bit binary number.

3) Modulo-6 Decoding (7400 NAND Gates): Since every die can have only 6 faces, it is important to make sure that the counter does not count beyond that number. A 7400 quad NAND gate network observes the QA–QD outputs and, when it sees the count getting into an invalid state, pushes the CLR/LOAD input of the counter, bringing the output count back into the 1-6 range.

4) Display Driving (CD4511 BCD-to-7-Segment Latch/Decoder): The CD4511 takes the BCD value from the counter, latches and decodes it, and drives the seven segment lines (a-g) of a common-cathode display. Current-limiting

resistors are placed in series with each segment to protect the display.

B. Circuit Setup

Components :

Breadboard

Resistors: 1k Ω , 10k Ω , 330 Ω

Capacitor: 0.1 μF

ICs used :

- NE555 Timer
- 74161 counter
- 7400 NAND
- CD4511 (BCD to Display decoder)

7 Segment Display

The circuit was developed using an analog and digital training system breadboard from the MCP M21-7000 series from the Analog company as illustrated in Fig. 2. The 555 timer section oscillates at a frequency that is determined by a 10 k Ω timing resistor and a 0.1 μF timing capacitor connected between the THRESHOLD/DISCHARGE pin and ground. The outputs of the decoder are connected in the BCD lines and are sent to the CD4511 whose outputs control the on-board seven-segment display via current limiting resistors.

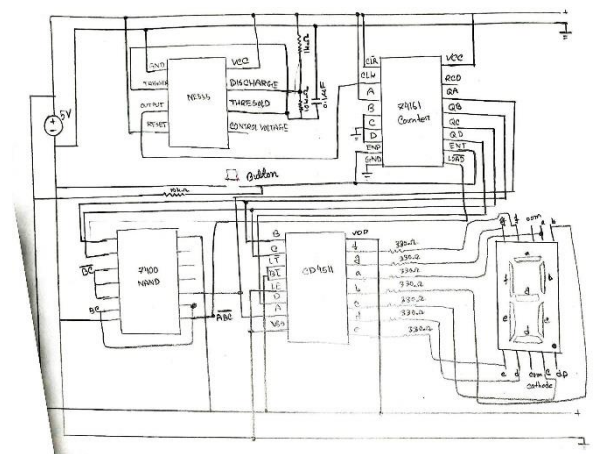


Fig. 1. Circuit schematic of the Electronic Dice Roller (555 timer, 74161 counter, 7400 NAND decoding, CD4511 driver, and 7-segment display).

C. Simulations / Testing

We confirm the 555 output frequency on the oscilloscope, and see the counter cycles from 1 to 6, reset without skipping or freezing when it counts for an invalid state, and see the seven-segment display with a readable digit for each valid count.

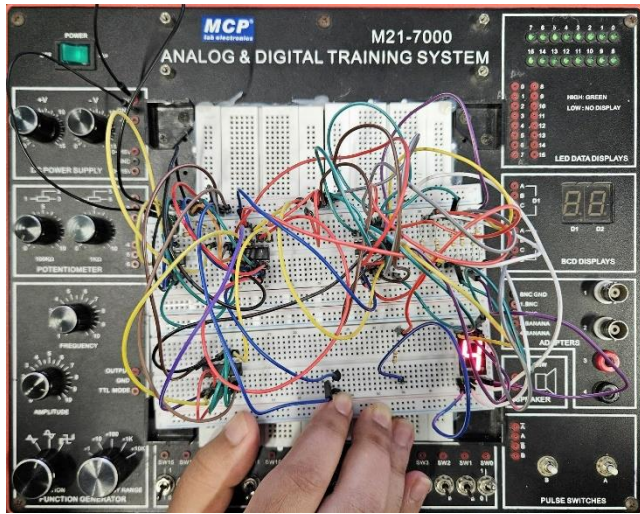


Fig. 2. Circuit implementation on the MCP M21-7000 training system breadboard

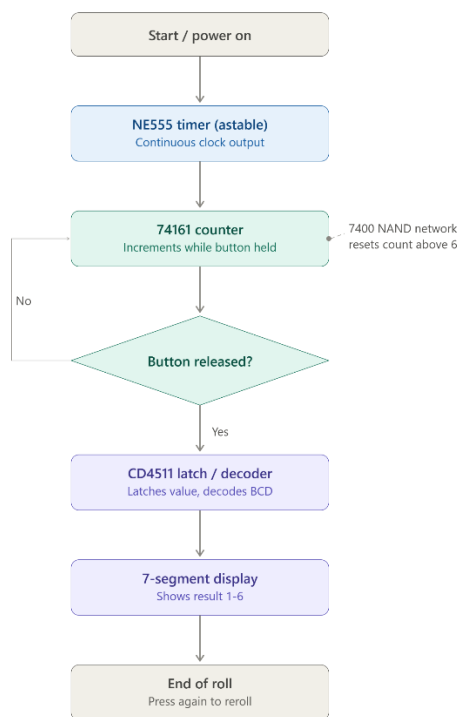


Fig. 3. Project Flowchart

D. Challenges

Firstly it was very hard to connect the wires to the right pin of the IC. But when we build our first circuit it was giving us the value between 4-6. Then we changed the capacitor and

resistance, then it was showing value between 3-6. Then we thought something was missing in the logic. Then we set frequency by the 10k Ω resistor and 0.1 μ F capacitor, then we got our desired values 1-6. Now when we press the button of the circuit it generates random value between 1-6. Push-button debounce causing multiple counts per press, correctly tuning the NAND decode network so the counter resets exactly at the right count, wiring density on the breadboard (Fig. 2) making faults hard to trace, and ensuring the CD4511's LE/BI/LT control lines were held in the correct state for norm

II. Conclusion

Designed and tested a 555 timer, 74LS190 counter, 74LS10 NAND gate and 4511 decoder based discrete logic electronic dice (no microcontroller). Powered by a free running clock, with the manual enable/freeze control, Circuit cycles the digits 1-6 on a 7-segment display. Real problems found and solved - incorrect preset numbers, resulting in false number range; ripple-counter glitch, blanking the display; and control-pin (BI/LT/LE) wiring errors. Says that combinational logic (reset condition using NAND), sequential logic (counter state) and BCD to 7 segment decoding co-operating.

III. References

- [1] TEXAS INSTRUMENTS, *xx555 PRECISION TIMERS*, NE555 DATASHEET, SLFS022, SEP. 1973 [REVISED 2026]. [ONLINE]. AVAILABLE: [HTTPS://WWW.TI.COM/LIT/DS/SYMLINK/NE555.PDF](https://www.ti.com/lit/ds/symlink/ne555.pdf)
- [2] TEXAS INSTRUMENTS, *SYNCHRONOUS 4-BIT COUNTERS*, SN74LS161A DATASHEET, SDLS060, OCT. 1976 [REVISED MAR. 1988]. [ONLINE]. AVAILABLE: [HTTPS://WWW.TI.COM/LIT/DS/SYMLINK/SN74LS161A.PDF](https://www.ti.com/lit/ds/symlink/sn74ls161a.pdf)
- [3] TEXAS INSTRUMENTS, *QUADRUPLE 2-INPUT POSITIVE-NAND GATES*, SN7400 DATASHEET. [ONLINE]. AVAILABLE: [HTTPS://WWW.TI.COM/LIT/DS/SYMLINK/SN7400.PDF](https://www.ti.com/lit/ds/symlink/sn7400.pdf)
- [4] TEXAS INSTRUMENTS, *CMOS BCD-TO-7-SEGMENT LATCH DECODER DRIVERS*, CD4511B DATASHEET. [ONLINE]. AVAILABLE: [HTTPS://WWW.TI.COM/LIT/DS/SYMLINK/CD4511B.PDF](https://www.ti.com/lit/ds/symlink/cd4511b.pdf)
- [5] ELECTRONICS TUTORIALS, "MULTIVIBRATORS INCLUDING MONOSTABLE, ASTABLE AND BISTABLE," ELECTRONICS-TUTORIALS.WS. [ONLINE]. AVAILABLE: [HTTPS://WWW.ELECTRONICS-TUTORIALS.WS/SEQUENTIAL/SEQ_3.HTML](https://www.electronics-tutorials.ws/sequential/seq_3.html)
- [6] M. M. MANO, *DIGITAL DESIGN*, 5TH ED. UPPER SADDLE RIVER, NJ, USA: PEARSON, 2013.